

Mathaus Krüger

AI Developer · Agentic AI · RAG Systems · LLM Applications

Pelotas, Brazil – Remote

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Professional Summary

AI developer focused on building LLM-powered systems that augment human cognition rather than replace human interaction.

Creator of **2Human**, a conversational AI agent grounded in Schema Therapy literature, combining RAG architecture, semantic knowledge retrieval, and persistent emotional memory for personalized user interaction.

Background in computational biology, machine learning research, and QA automation. Interested in developing AI systems for psychological self-reflection, cognitive support, and human-AI collaboration.

Projects

2Human — Conversational AI Agent

2human.streamlit.app
github.com/mathauskruger/2human-alpha

Conversational AI system designed to support emotional reflection and pattern recognition using Schema Therapy knowledge.

Key features:

- RAG pipeline built on clinical literature (Young, Klosko, Weishaar)
 - FAISS vector database with 2000+ indexed semantic knowledge chunks
 - Persistent per-user emotional memory stored in structured JSON
 - Schema detection system requiring multiple evidence signals before classification
 - Bilingual interaction (English / Portuguese)
 - Deployment using Streamlit Cloud and OpenRouter LLM APIs
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LifeSchema — Schema Therapy Chatbot

Prototype system integrating psychological self-assessment with conversational AI.

Key features:

- Schema therapy questionnaire and scoring system
 - Radar chart visualization of schema patterns
 - Supabase authentication and user persistence
 - AI response personalization based on assessment results
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Neuropsychological Screening Agent (in development)

Conversational AI system designed for **preliminary adult giftedness screening (AHSD)**.

Key features:

- Structured conversational interview flow
 - RAG knowledge base based on neuropsychology literature
 - Longitudinal pattern analysis of cognitive indicators
 - Automated report generation for preliminary screening insights
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Professional Experience

Research Assistant — Machine Learning / Biomedical Data

UCPel. Aug 2023 – Jan 2024

- Applied graph-based semi-supervised learning to biomedical datasets using Python
 - Assisted in analysis of complex biological networks and pattern detection
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QA Automation Intern

Compass UOL. May 2023 – Aug 2023

- Built automated API testing suites using RestAssured
 - Developed automated validation pipelines for backend services
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Research Scholar — Computational Biology

UFPel. 2019 – 2022

- Conducted computational analysis of neurobiological datasets
 - Applied programming and statistical methods to biological systems research
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Education

Bachelor of Science in Biological Sciences / Bioinformatics
Federal University of Pelotas (UFPeI)
2016 – 2022

Technical Skills

Agentic AI Development
RAG Systems
LLM Integration
Prompt Engineering
Vector Databases
API Integration
System Design
Test Automation

Tech Stack

Python
LangChain
FAISS
OpenAI API
Gemini API
OpenRouter
Streamlit
Supabase
REST APIs
JSON
Git / GitHub

Languages

Portuguese — Native
English — Fluent
Spanish — Basic